

Moist V1 status report: the parameter-sensitivity program and the gross-moist-stability regime boundary

2026-07-19

Summary

Moist V1 (a passive column-water-vapor field W riding on the dry Sobel–Schneider circulation) is implemented, validated, and its full placeholder-parameter sensitivity program is complete: four scans covering the eddy moisture diffusivity D , the precipitation threshold W_c , the convective timescale τ_c , and the stratification depth Δ_z , eighteen steady-state runs in all. One result organizes everything (Fig. 1): because the equilibrium overturning is weak, W is nearly uniform at the value $W_q = W_c + \tau_c E_0$ set by the local evaporation–precipitation balance, so the sign of the gross moist stability $\hat{H} = \hat{S} - L_v(2a-1)W$, and with it the direction of the mean meridional MSE flux, is decided by a single comparison: W_q against $W^* = \hat{S}/(L_v(2a-1))$. Every crossover this comparison predicts in closed form is confirmed by the runs. At the boundary the mean MSE flux vanishes by exact cancellation of its dry-static and latent parts, leaving the diagnostic eddy flux $-L_v D \partial_y W$ as the only meridional MSE transport: the regime the eddy-diffusion program is aimed at, now locatable exactly. Remaining before V2: the ERA5 calibration of a and D (Spencer), then latent heating.

1 Where the moist model stands

V1 adds one prognostic equation to the dry model: column water vapor $W(y, t)$, transported by the lower-branch flow and mixed down-gradient, with uniform evaporation E_0 and a quasi-equilibrium precipitation sink $P = (W - W_c)^+/\tau_c$ (Appendix A has the full equation set). The coupling is strictly one-way: W feeds back on nothing, and the dry fields of a moist run are bit-for-bit identical to a dry twin, a tested invariant that also holds across every run discussed here. V2 will close the loop by letting P heat the temperature equation.

Implementation status: merged to `main`, 338 tests, with exact invariants (moisture mass conservation to roundoff, mirror parity, bit-exact restarts, dry invariance) and a regression baseline. The moist parameters are placeholders pending an ERA5 calibration of the lower-layer moisture fraction a and the diffusivity D ; that calibration is the only open pre-V2 task besides V2 itself.

The sensitivity program ran on the production dry formulation (staggered grid, EMFD gate on, MC-limited stencil; `ny`=801, $\Delta t = 30$ s, equinoctial $\sin^2$ forcing) with every run integrated to a slow-drift-gated statistical steady state (gate firings: day 1772–1896). The four scans:

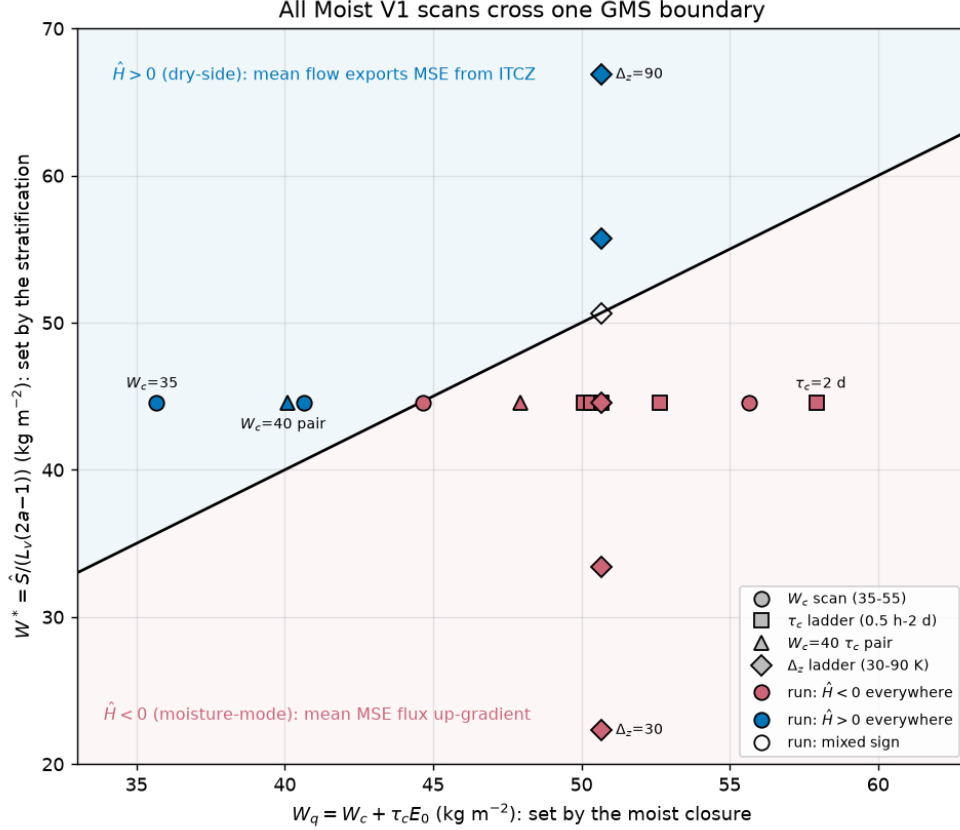


Figure 1: Every scanned run placed by what its parameters set: the quiescent plateau $W_q = W_c + \tau_c E_0$ (the W the moist closure relaxes to where the circulation is weak) against the crossover $W^* = \hat{S}/(L_v(2a-1))$ (the W at which $\hat{H}$ changes sign, proportional to Δ_z). Fill color gives the measured sign of $\hat{H}(y)$ over the whole domain. Runs sit on the predicted side of the diagonal in every case; the one mixed-sign run ($\Delta_z=68.2$ K) sits on it. The D ladder collapses onto the shared center point: D moves neither coordinate.

scan	values	runs	center point
D	$\{0, 1, 2\} \times 10^6 \text{ m}^2/\text{s}$	3	shared: the default run
W_c	35, 40, 44, 50, 55 kg/m^2	4 (+center)	($W_c=50$)
τ_c	0.5, 2, 4, 16, 48 h; pair at $W_c=40$	6 (+center)	($\tau_c=4$ h)
Δ_z	30, 45, 60, 68.2, 75, 90 K	5 (+center)	($\Delta_z=60$ K)

All other parameters sit at their defaults (Appendix B); the default run ($W_c=50$, $\tau_c=4$ h, $\Delta_z=60$ K, $D=10^6$) is the shared center of every ladder.

2 One boundary organizes all four dials

The equilibrium overturning is weak because it is eddy-driven: the zonal momentum balance is $\beta y v \approx \text{EMFD}$ (drag is two orders down), giving $|v|_{\max} = 0.37 \text{ m/s}$ at the default $v_d = 2.5 \text{ m/s}$. Weak transport means the moisture equation is nearly in local balance everywhere: $P \approx E_0$, so W

sits within 0.01–1.4 kg/m² of the *quiescent plateau*

$$W_q = W_c + \tau_c E_0, \quad (1)$$

the value at which the precipitation closure removes exactly what evaporation supplies. (Measured: plateau matches this to $\leq 7.5 \times 10^{-8}$ kg/m² in every run.)

With W pinned near W_q , the gross moist stability $\hat{H}(y) = \hat{S} - L_v(2a-1)W(y)$ has, to the accuracy of those small departures, one sign for the whole domain, decided by W_q against

$$W^* = \frac{\hat{S}}{L_v(2a-1)} = \frac{C \delta \Delta_z / H}{L_v(2a-1)} = 44.57 \text{ kg/m}^2 \times \frac{\Delta_z}{60 \text{ K}}. \quad (2)$$

$\hat{H} > 0$ ($W_q < W^*$): the mean overturning exports MSE from the ITCZ, the familiar dry-side regime. $\hat{H} < 0$: the mean MSE flux is up-gradient (into the ITCZ), the moisture-mode side. Figure 1 shows the runs in the (W_q, W^*) plane; the diagonal is the regime boundary, and each dial moves the system along exactly one axis: W_c and τ_c move W_q , Δ_z moves W^* , D moves neither.

Three crossover predictions follow, and the runs confirm each (all numbers are $\hat{H}$ at the ITCZ in MJ/m²):

- **In W_c at fixed τ_c :** $W_c^* = W^* - \tau_c E_0 = 43.91$. Measured ladder +15.5, +6.7, −0.3, −10.8, −19.6 across $W_c = 35 \dots 55$; the $W_c=44$ run is in effect the boundary run.
- **In Δ_z :** $\Delta_z^* = 68.20$ K at the default W_q . The run placed at 68.2 K lands mixed-sign with $\hat{H}(0) = -0.16$.
- **In τ_c :** only reachable if $W_c < W^*$, at $\tau_c^* = (W^* - W_c)/E_0$. At $W_c=40$ that is 9.94×10^4 s (1.15 d), and the run pair at $\tau_c = 0.5 \text{ h} / 2 \text{ d}$ brackets it: $\hat{H}(0) = +7.84 \rightarrow -7.88$. At the default $W_c=50$ the plateau starts above W^* , so shrinking τ_c only saturates $\hat{H}(0)$ at $\hat{S} - L_v(2a-1)W_c = -9.5$; no τ_c crosses.

At the boundary the eddy flux is the only MSE transport. The mean flux $v\hat{H}$ is the residual of two large opposing parts, the poleward dry-static flux $\hat{S}v$ and the equatorward latent flux $-L_v(2a-1)Wv$. At $\Delta_z=68.2$ these are +30.18 and −30.18 MW/m at the flux latitude: the net is 0.00, and the diagnostic eddy flux (0.13 MW/m) carries everything. The W_c scan found the same at its crossover (eddy twice the net). Away from the boundary the mean part dominates by factors of 20–60.

The mixed-sign window is as narrow as the W field is flat. The spatial spread of $\hat{H}$ is $L_v(2a-1)$ times the spatial spread of W (≈ 0.17 kg/m² at default τ_c), about 0.3 MJ/m². Turning any dial slides the nearly flat $\hat{H}(y)$ rigidly through zero, so both signs coexist over only ~ 0.17 kg/m² of W_c or ~ 0.23 K of Δ_z : the domain changes regime almost as a unit. The window widens in proportion to the transport departures, hence linearly in τ_c (~ 3 K of Δ_z at $\tau_c=2$ d).

Finally, every scanned configuration precipitates everywhere ($P_{\min} \geq 3.4$ mm/d against $E_0 = 3.97$ mm/d): with uniform evaporation and departures this small, W never falls below W_c anywhere. A subtropical dry zone would need either structured evaporation or a much stronger circulation.

3 The convective timescale τ_c

τ_c has one trivial role and one dynamical role.

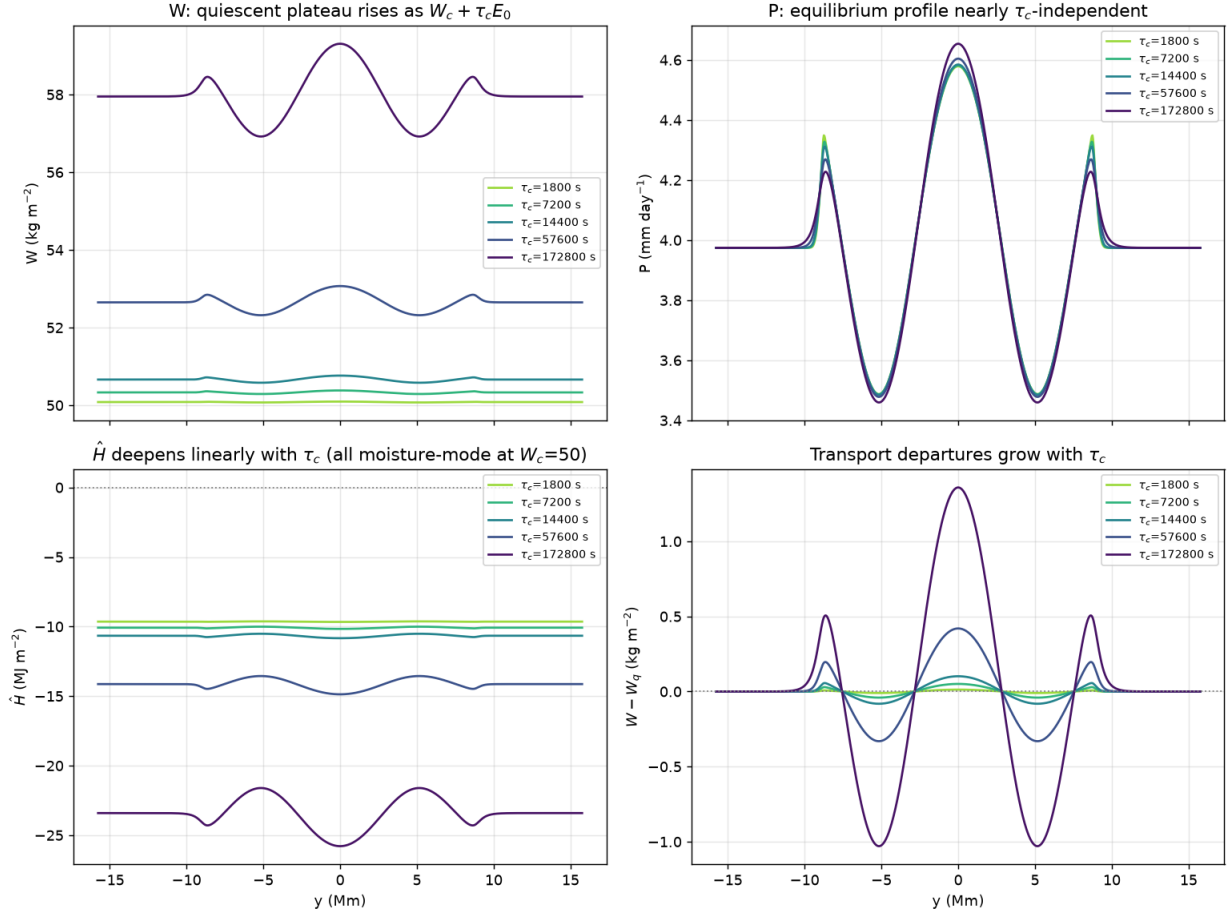


Figure 2: τ_c ladder at $W_c=50$ (green $\rightarrow$ purple: 0.5 h $\rightarrow$ 2 d). Clockwise from top left: W (the plateau climbs as $W_c + \tau_c E_0$); P (nearly unchanged: it is fixed by the moisture budget); departures $W - W_q$ (amplitude grows with τ_c , shape fixed); $\hat{H}$ (deepens, no sign change at this W_c).

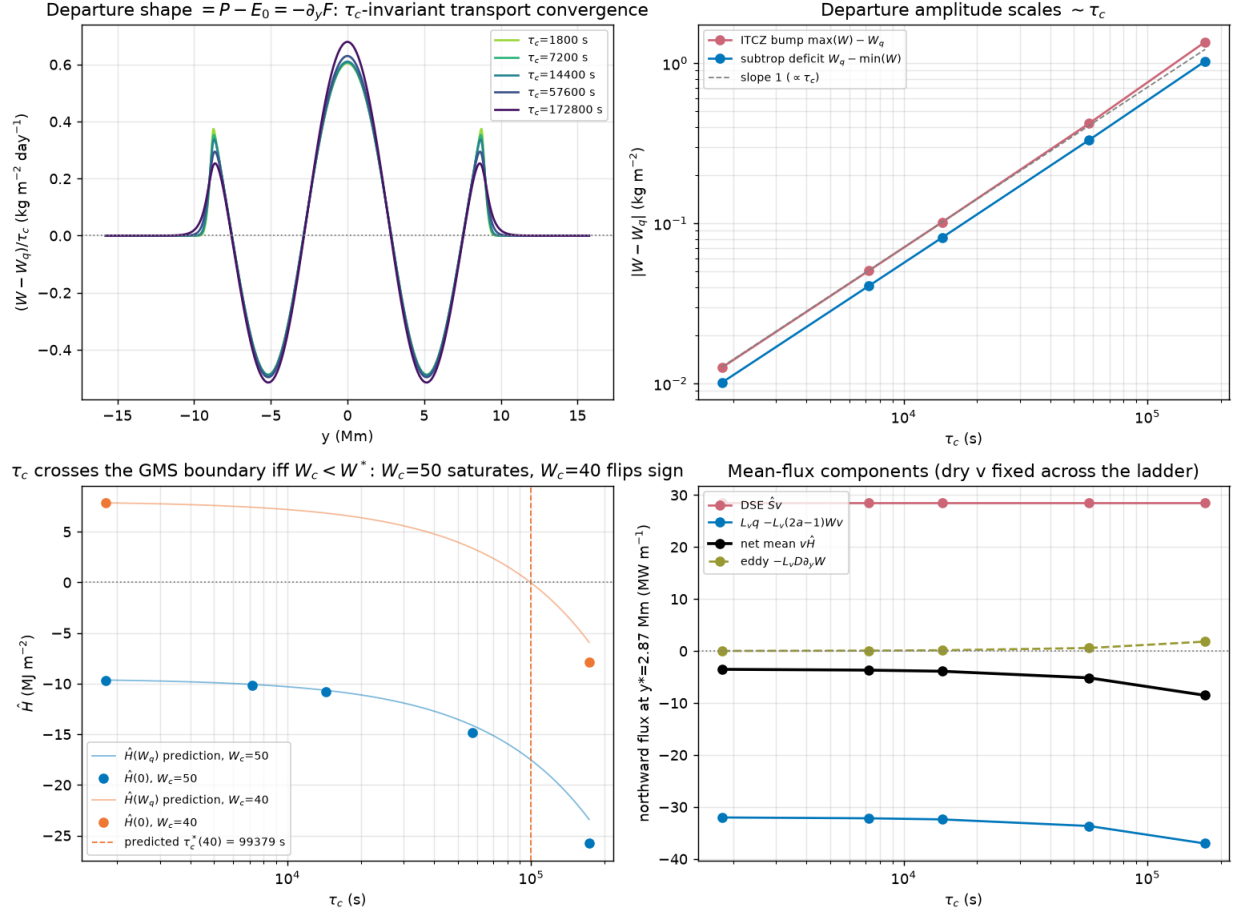


Figure 3: τ_c summary. Top left: the departure shape $(W - W_q)/\tau_c = P - E_0 = -\partial_y F$ collapses across the ladder: the transport convergence is τ_c -invariant while the departure amplitude scales with τ_c . Top right: that amplitude scaling (log-log, slope-1 guide). Bottom left: $\hat{H}(0)$ against τ_c for $W_c=50$ (saturates) and $W_c=40$ (crosses zero at the predicted τ_c^* , dashed); points sit below the plateau-based curves by the ITCZ-bump contribution $L_v(2a-1)\Delta W$. Bottom right: mean-flux decomposition at the flux latitude.

Trivial: it sets the plateau. A column raining at rate P must hold $P\tau_c$ of water above threshold, so where $P = E_0$ the column holds $W_c + \tau_c E_0$. Slower convective removal means a moister column at the same rain rate. This is a single-point balance with no dynamics in it; the runs track it to $\leq 7.5 \times 10^{-8} \text{ kg/m}^2$ over a $\times 96$ range of τ_c (0.5 h to 2 d).

Dynamical: it is the gain on the transport structure. At steady state $(W - W_q)/\tau_c = P - E_0 = -\partial_y F$ exactly, where F is the total meridional moisture flux. The measured departure profiles collapse under this scaling (Fig. 3, top left): the moisture-flux convergence is set by the dry circulation and the nearly fixed W , and is therefore τ_c -invariant, while the W departures needed to convert that convergence into extra or reduced rain scale with τ_c . Measured: the ITCZ bump grows $0.013 \rightarrow 1.359 \text{ kg/m}^2$ across the ladder, with $(\max W - W_q)/\tau_c$ constant to within 11% (the excess tracks the plateau's own growth, since $F \propto W$). Equilibrium P itself moves by at most 0.09 mm/d across the whole ladder. Consequences: $\hat{H}(0)$ deepens linearly, $-9.7 \rightarrow -25.8 \text{ MJ/m}^2$ at $W_c=50$, and the mixed-sign window of Sec. 2 widens linearly.

Regime reach. As shown in Sec. 2: crossing in τ_c requires $W_c < W^*$ (confirmed at $W_c=40$); at the default $W_c=50$ the dial saturates on the moisture-mode side.

Numerics. D and P are stepped at the lagged $n-1$ level, so the precipitation stability bound is $2\Delta t/\tau_c \leq 1$, i.e. $\tau_c \geq 60 \text{ s}$ at $\Delta t=30$. Ten-day smokes at $\tau_c = 60, 120, 300 \text{ s}$ (bound ratios 1.0, 0.5, 0.2) all ran clean, so the bound is not a practical constraint for any physical τ_c . The dry fields of every τ_c run (including the $W_c=40$ pair) are bit-identical to the center run ($\max |\Delta u| = 0$ over all 1855 daily fields; all gate day 1855), as passivity requires.

4 The stratification depth Δ_z

Δ_z is the one dial with a dual role: it sits in the dry thermodynamic equation (the adiabatic term $(\delta\Delta_z/H)\partial_y v$), so it changes the circulation itself, and it sets the gross dry stability $\hat{S} = C\delta\Delta_z/H$, so it moves the crossover W^* proportionally. The moist closure does not see it: the plateau stays at $W_q = 50.66$ across the whole ladder (measured ITCZ W : 50.74–50.83 over a $\times 3$ range of Δ_z).

Energetics. $\hat{H}(0)$ runs from -49.9 MJ/m^2 at 30 K to $+28.2$ at 90 K, crossing zero at the predicted 68.20 K; the mean MSE flux at the flux latitude runs $-26.5 \rightarrow +8.3 \text{ MW/m}$, reversing there. The DSE component grows only 63% ($20.9 \rightarrow 34.1 \text{ MW/m}$) over a $\times 3$ growth in $\hat{S}$ because the circulation weakens at the same time; the latent component shrinks with $|v|$ ($-47.4 \rightarrow -25.8 \text{ MW/m}$) at fixed W .

Dry response. Over $30 \rightarrow 90 \text{ K}$ the jet weakens $31.9 \rightarrow 25.9 \text{ m/s}$ (-19%) and $|v|_{\max}$ falls $0.53 \rightarrow 0.29 \text{ m/s}$ (-45%), with the flux latitude moving poleward ($2.56 \rightarrow 3.03 \text{ Mm}$) and the convergence gate firing progressively later (day 1772 $\rightarrow$ 1896). Mechanism: the steady thermodynamic balance $(\delta\Delta_z/H)\partial_y v = (\theta_E - \theta)/\tau$ requires less divergence per unit diabatic heating when the stratification is deeper, so v weakens with Δ_z . The measured response is shallower than the fixed-heating $1/\Delta_z$ scaling ($\times 1.8$ over a $\times 3$ range) because the disequilibrium $\theta_E - \theta$ adjusts as well.

5 The other two dials: W_c and D

The W_c and D scans (runs and figures in `model_output/moist_v1_wc_scan/` and `model_output/moist_v1_validation/`):

W_c (**35–55**). Shifts W_q one-for-one. The $\hat{H}(0)$ ladder runs from $+15.5$ to -19.6 MJ/m^2 across

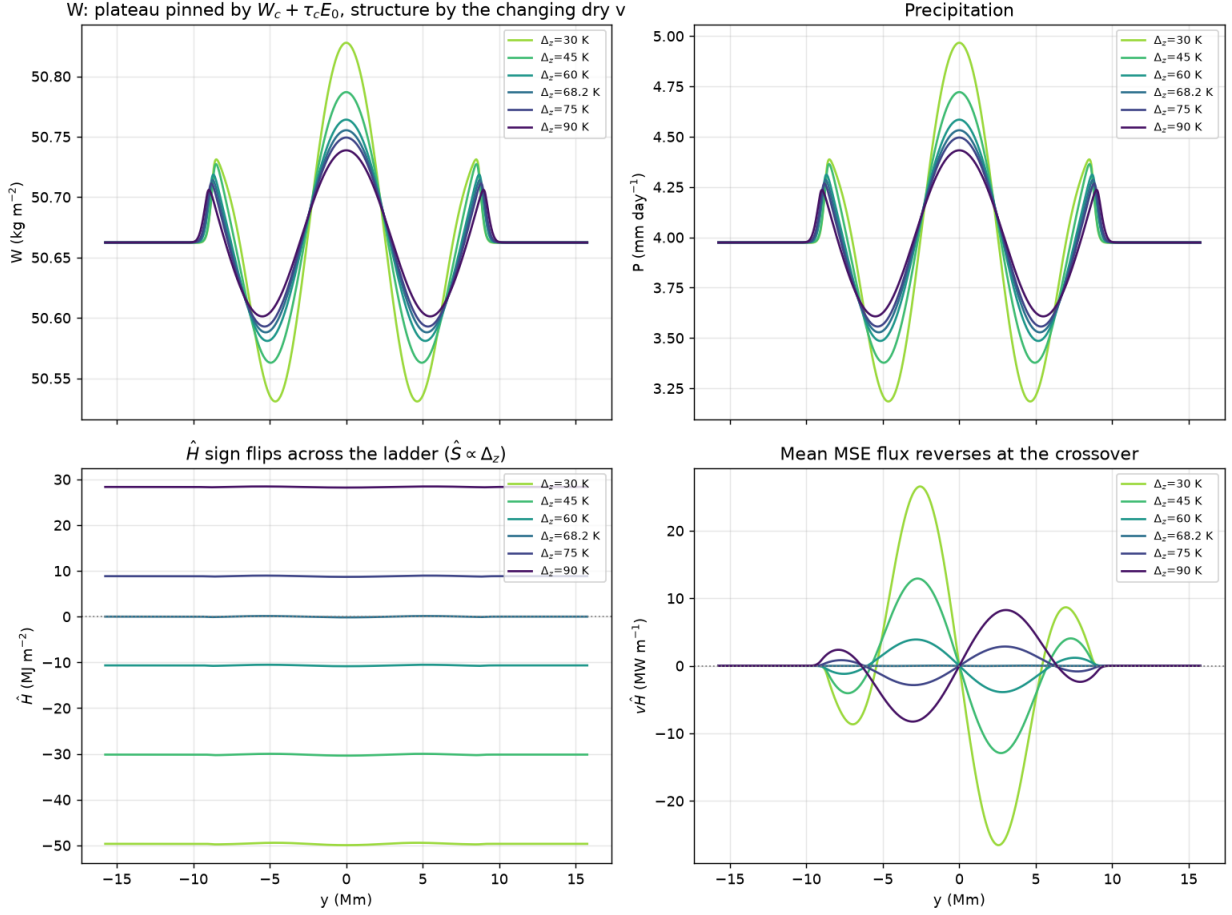


Figure 4: Δ_z ladder (green $\rightarrow$ purple: 30 $\rightarrow$ 90 K). W and P keep the same plateau (the moist closure does not see Δ_z) with structure amplitude following the weakening dry v ; $\hat{H}$ slides through zero; the mean MSE flux $v\hat{H}$ reverses direction across the ladder.

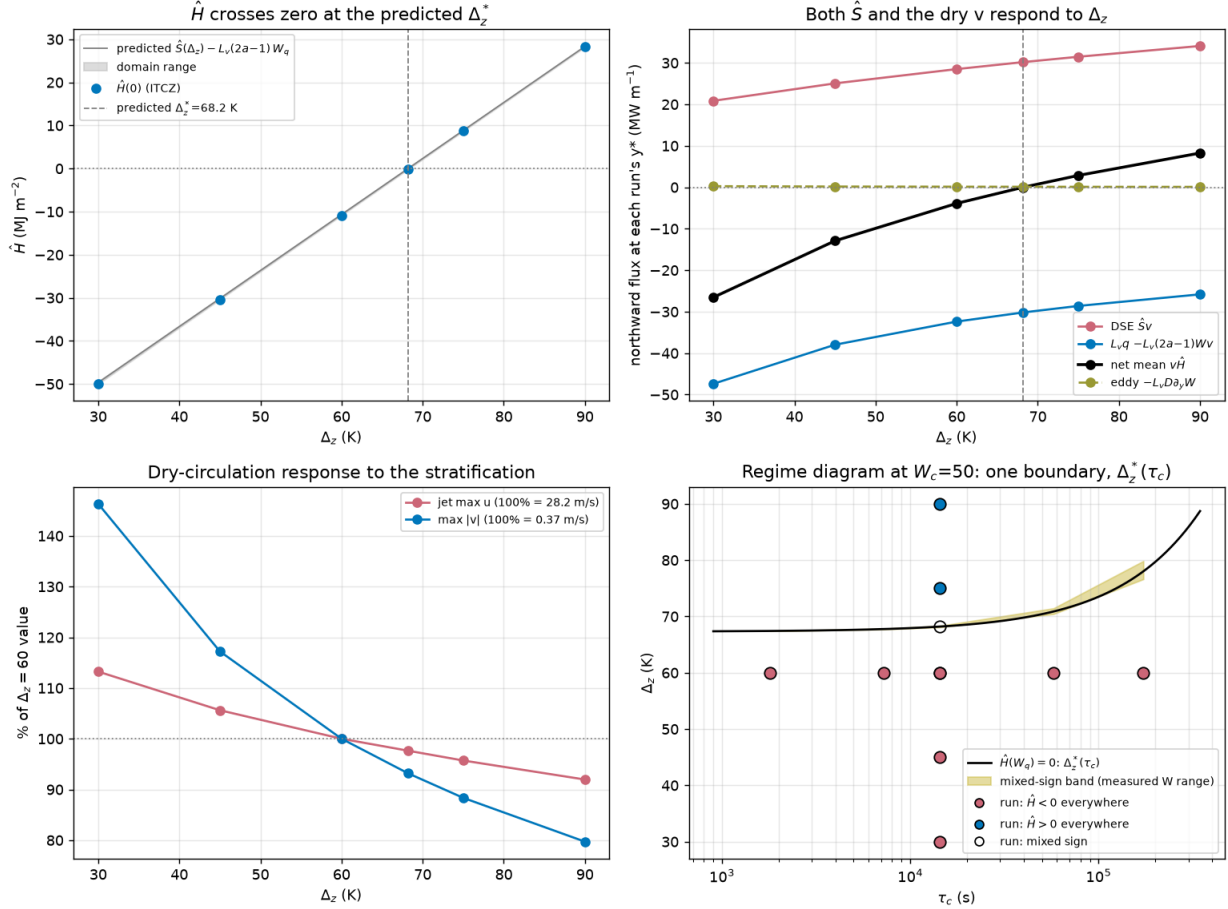


Figure 5: Δ_z summary. Top left: $\hat{H}(0)$ crosses zero at the predicted $\Delta_z^* = 68.2$ K. Top right: flux decomposition; the DSE flux rises with $\hat{S}$ but sublinearly (the circulation weakens), the latent flux shrinks with v , the net reverses at the crossover, the eddy flux stays at 0.1–0.25 MW/m throughout. Bottom left: dry-circulation response (percent of the $\Delta_z=60$ values). Bottom right: the regime diagram at $W_c=50$ in the (τ_c, Δ_z) plane with the boundary $\Delta_z^*(\tau_c)$ and the mixed-sign band from the measured W ranges.

$W_c = 35 \dots 55$, crossing at $W_c^* = 43.91$; the $W_c=44$ run sits at the boundary ($\hat{H}(0) = -0.32$, net mean flux -0.06 MW/m, eddy flux 0.12 , i.e. twice the net).

D (**0, 1, 2** $\times 10^6$ m²/s). In V1, D is the eddy-flux dial and nothing else: the eddy MSE flux scales with it (0, 0.28, 0.46 MW/m), while its leverage on W itself is negligible ($\max |W_{D2} - W_{D0}| = 0.015$ kg/m²), because the departures D would smooth are already tiny. D moves neither coordinate of Fig. 1.

6 Implications for V2

dial	sets	crossover	independent leverage
W_c	W_q level	$W_c^* = W^* - \tau_c E_0 = 43.91$	none beyond W_q
τ_c	W_q offset $\tau_c E_0$	$\tau_c^* = (W^* - W_c)/E_0$, iff $W_c < W^*$	departure amplitude, window width
Δ_z	W^* ($\propto$)	$\Delta_z^* = 68.20$ K at default W_q	dry circulation strength
D	(neither)	none	eddy MSE flux magnitude
v_d	(neither)	none	overturning $\Rightarrow$ departures, mean flux

1. **Regime placement is now a closed-form choice.** To start V2 in a chosen GMS regime, pick $W_q - W^*$ directly; the boundary-adjacent states already exist as runs with restart files ($W_c=44$ and $\Delta_z=68.2$).
2. W_c and τ_c are degenerate for the regime. Only their combination $W_q = W_c + \tau_c E_0$ decides the sign of $\hat{H}$; they are not independent regime dials. τ_c 's independent role is the departure amplitude (and, in V2, the stiffness of the precipitation coupling).
3. **V1's energetic blandness at the defaults is placement, not structure.** At the default parameters the system sits ~ 6 kg/m² deep on the moisture-mode side, with the mean flux dominating the eddy flux ~ 25 -fold. At the boundary that inverts: the eddy flux, the object the ERA5-calibrated D is meant to represent, becomes the entire MSE transport.
4. **Expect the picture to deform under V2 coupling.** Everything here is one-way; V2's latent heating lets P anomalies drive the circulation. On the $\hat{H} < 0$ side that closes a positive feedback loop (moisture-mode amplification), so the whole-domain-flips-as-a-unit behavior and the narrow mixed-sign window of Sec. 2 need not survive coupling [speculative]. The near-boundary states are the natural V2 starting points precisely because there the mean-flow feedback is weakest. The V2 gate decision (kinematic $\mathcal{H}(\partial_y v)$, validated dry) is already made.
5. **Costs.** A gated moist steady-state run is ~ 1855 days, measured at 27–45 min on the numpy backend (~ 0.9 – 1.4 s/day across contention states); the numba kernel does not yet cover moisture (deferred until V1 physics froze, which it now has).

A Model equations

Upper-layer fields $u(y, t)$, $v(y, t)$, $\theta(y, t)$ on an equatorial β -plane channel $|y| \leq L$, walls at $\pm L$; moist V1 adds $W(y, t)$. $\mathcal{H}$ is the Heaviside function.

Zonal momentum (SS09 Eq. 2.1 with the thermodynamic gate):

$$\frac{\partial u}{\partial t} = v(\beta y - \partial_y u) - \mathcal{H}(\theta_E - \theta)(\partial_y v)u - \varepsilon_u u - \underbrace{v_d \mathcal{H}(u) \operatorname{sgn}(y) \partial_y u}_{\text{EMFD}}. \quad (3)$$

The second term is vertical exchange with the lower layer where convection is active (ascending air carries zero zonal momentum); the gate argument is $\theta_E - \theta$ where SS09 use $\mathcal{H}(\partial_y v)$, a deviation that suppresses a grid-scale artifact of the original collocated grid (the two arguments coincide in steady state through the thermodynamic balance; the kinematic gate is validated on the staggered grid and will land with V2). The EMFD (SS09 Eq. 2.5) represents extratropical eddies depositing easterly momentum inside the westerly Hadley cell; it acts as poleward advection of u at speed v_d where $u > 0$.

Meridional momentum (SS13 corrigendum form; the factor 2 is intentional):

$$2\frac{\partial v}{\partial t} + \beta y u = -\frac{gH}{T_0} \partial_y T + k_v \partial_y^2 v, \quad T = \theta (p_t/p_s)^{R/c_p} = \theta/1.6. \quad (4)$$

Thermodynamic:

$$\frac{\partial \theta}{\partial t} + \frac{\delta \Delta_z}{H} \partial_y v = \frac{\theta_E - \theta}{\tau} \left(+ \kappa_\theta \partial_y^2 \theta, \kappa_\theta = 0 \text{ by default} \right), \quad (5)$$

with the equinoctial $\sin^2$ radiative-convective profile

$$\theta_E(y) = \theta_{00} - \Delta_y \sin^2\left(\frac{\pi y}{2y_1}\right) \quad (|y| < y_1), \quad \theta_E = \theta_{00} - \Delta_y \quad (|y| \geq y_1). \quad (6)$$

Moisture (V1, passive):

$$\frac{\partial W}{\partial t} + \partial_y \left[\underbrace{-(2a-1)vW}_{\text{mean transport}} - \underbrace{D\partial_y W}_{\text{eddy mixing}} \right] = E_0 - P, \quad P = \frac{(W - W_c)^+}{\tau_c}. \quad (7)$$

The transport velocity is $-(2a-1)v$: the lower branch moves opposite the upper-layer v , and with fraction a of the column water in the lower layer the column-integrated flux reduces to this coefficient. Nothing feeds back on u, v, θ .

Column energetics (derivations in docs/moist_v1_mse_budget.pdf): with column heat capacity $C = c_p \Pi \Delta p / g$ and $L_v = C \Lambda_{\text{conv}}$,

$$\hat{S} = C \frac{\delta \Delta_z}{H}, \quad \hat{H} = \hat{S} - L_v(2a-1)W, \quad W^* = \frac{\hat{S}}{L_v(2a-1)}, \quad (8)$$

and the steady column-MSE budget carries the three fluxes used throughout this report: mean DSE $\hat{S}v$, mean latent $-L_v(2a-1)Wv$ (their sum is the net mean flux $v\hat{H}$), and eddy $-L_v D \partial_y W$.

Numerics. Arakawa-C staggered grid (v on the $ny-1$ interior faces; u, θ, W on centers); leapfrog with Asselin filter 0.04; upwind meridional advection of u ; MC-limited one-sided EMFD stencil (the gate-on production formulation); W transport in finite-volume flux form with MUSCL-MC face reconstruction (conserves total water to roundoff); the moist diffusion D and sink P stepped at the lagged $n-1$ level (stability bounds $2D\Delta t/\Delta y^2 \leq 0.5$, $2\Delta t/\tau_c \leq 1$). Steady-state stopping: KE and θ -variance convergence plus the slow-drift gate (jet latitude, $|v|_{\text{max}}$, equatorial disequilibrium each flat to 0.2% over a drag-timescale window).

B Parameter values

Dry defaults (the Zhang et al. 2025 lineage; k_v is 100× SS09’s printed value, following their code):

symbol	value		meaning
β	2×10^{-11}	$\text{m}^{-1}\text{s}^{-1}$	planetary vorticity gradient
ε_u	10^{-8}	s^{-1}	Rayleigh drag on u
v_d	2.5	m/s	EMFD coefficient
k_v	7.786×10^5	m^2/s	eddy viscosity on v
τ	37	d	Newtonian cooling timescale
H	16 000	m	tropopause height
δ	4 000	m	upper-layer thickness
Δ_z	60	K	vertical stability (scanned 30–90)
T_0	300	K	reference temperature
θ_{00}	330	K	background θ_E
Δ_y	50	K	equator–pole θ_E contrast
y_1	9.439×10^6	m	forcing half-width
L	1.5751×10^7	m	channel half-width
g	9.81	m/s^2	gravity

Moist V1 (**placeholders** pending the ERA5 calibration of a and D) and column constants:

symbol	value		meaning
a	0.85		column-water-vapor fraction in the lower layer
D	10^6	m^2/s	eddy moisture diffusivity (scanned $0\text{--}2 \times 10^6$)
W_c	50	kg/m^2	precipitation threshold (scanned 35–55)
τ_c	14 400	s	convective timescale (scanned 1 800–172 800)
E_0	4.6×10^{-5}	$\text{kg m}^{-2}\text{s}^{-1}$	evaporation (= 3.97 mm/d)
C	5.2×10^6	$\text{J m}^{-2}\text{K}^{-1}$	column heat capacity
L_v	2.5×10^6	J/kg	latent heat (= $C\Lambda_{\text{conv}}$)

Numerics: $n_y = 801$ ($\Delta y = 39.4$ km), $\Delta t = 30$ s, Asselin 0.04, numpy backend (numba does not yet cover moisture). Derived at the defaults:

$$\hat{S} = 7.8 \times 10^7 \text{ J/m}^2, \quad W^* = 44.57, \quad W_q = 50.66 \text{ kg/m}^2, \quad W_c^* = 43.91, \quad \Delta_z^* = 68.20 \text{ K}.$$

C Run inventory and regeneration

All eighteen runs use

```
run-sw-model --ny 801 --dt 30 --enable-moisture --dw 1e6
--stop-at-steady-state --slow-drift-gate
```

plus the dial flag(s) (`--w-crit`, `--tau-c`, `--delta-z`), each in its own output directory with its final restart file.

scan	runs (directory)	dial values	gate day
D ladder	moist_v1_validation/D{0,1,2}	$D = \{0,1,2\} \times 10^6$	1855
W_c scan	moist_v1_wc_scan/wc{35,40,44,55}	W_c	1855
τ_c ladder	moist_v1_tauc_dz/tc{1800,7200,57600,172800}	τ_c	1855
$W_c=40$ pair	moist_v1_tauc_dz/wc40tc{1800,172800}	W_c, τ_c	1855
Δ_z ladder	moist_v1_tauc_dz/dz{30,45,68p2,75,90}	Δ_z	1772–1896

(Directories relative to `model_output/`; every $\Delta_z=60$, $\tau_c=14\,400$, $W_c=50$ slot reuses `moist_v1_validation/D1`. The τ_c and W_c runs share the D1 dry solution bit-for-bit, so gate day 1855 is exact; the Δ_z gate days are 1772, 1823, 1876, 1882, 1896 for 30, 45, 68.2, 75, 90 K.)

Analysis scripts (repo `scripts/`; each regenerates its figures from the run directories): `moist_v1_analysis.py` (D ladder + budget figures), `moist_v1_wc_scan_analysis.py`, `moist_v1_tauc_dz_analysis.py` (both ladders here, the scorecards, Figs. 2–5), and `moist_v1_regime_figure.py` (Fig. 1). This report: `latexmk -pdf -output-directory=docs docs/moist_v1_sensitivity_report.tex` from the repo root.